Filters, PCA, or Autoencoders? Benchmarking Image Denoising Under Gaussian and Impulse Noise

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***A*bstract—Noise enters into images at the sensor, in storage and during transmission, and it makes edges and fine texture more difficult for any later recognition step. This paper compares three ways of cleaning it up on equal terms: local spatial filters (Gaussian, median and bilateral), a PCA subspace projection, and a small convolutional denoising autoencoder (CDAE) with 66,593 parameters. We test all of them on 500 Fashion-MNIST images corrupted by Gaussian noise (σ = 15, 25, 50) and by salt- and-pepper noise (5% and 10% of pixels), using PSNR, SSIM and MSE values, and the time taken per image. The CDAE gives the best SSIM in all five cases and the best PSNR for strong Gaussian noise and both impulse settings, for example 21.95 dB at σ = 50 against 19.19 dB for the best filter. A tuned bilateral filter is still ahead in PSNR when the noise is mild (28.35 dB at σ = 15), and there every other method ends up below the noisy input. In short, the best denoiser depends on the type and strength of the noise; a single learned model is more robust, but it gives up a little on very light noise.**

Keywords—image denoising, pattern recognition, bilateral filter, principal component analysis, denoising autoencoder, PSNR, SSIM, impulse noise.

I. INTRODUCTION

Most images begin as measurements which are subject to noise. The increase in gain in order to brighten a dark room when using a camera will increase the noise, and the number of photons measured when using a low dose X-ray will be smaller. Faulty connections or faulty memory cells can cause noise as pixels may go white or black. All that is seen by the recognition system, however, is the noisy image.

The task of cleaning an image is an ill-posed problem. There are many different images that could give rise to the observed corrupted one, and any noise reduction technique will also destroy some genuine edges. Researchers have worked on the problem in about three ways. The first approach is to use a predetermined filter that acts locally on the image, such as Gaussian, median, or bilateral filters [1], [2]. They are computationally fast and require no training data, but they make only one assumption about images – they are locally smooth. The second is to use PCA to learn a low-dimensional representation of clean images [3], and throw away the dimensions containing mainly noise. The third is to train a neural net to predict the clean images from the corrupted ones [4], [5].

New algorithms are typically compared against only several other algorithms and noise type. That does not help much in case of choosing an algorithm not knowing what kind of noise will be used. That is why we conducted all three families of algorithms using one protocol: identical data splitting, identical noise generator, identical metrics and identical machine. Our contributions include:

- reproducible benchmark of six restoration algorithms under five noise types, including both Gaussian and impulse noise;
- tuning of classical algorithms based on validation technique and tuning of the number of PCA components separately;
- compact CDAE, trained on noise mixture, with cost evaluation compared to classical algorithms; and
- a study on situations where each of the algorithms helps, as well as when it decreases PSNR score below that of noisy signal.

In Section II we review the related work, in Section III we describe the data, noise models and methods used. Results and their discussion are presented in Section IV and conclusions are made in Section V.

II. LITERATURE SURVEY

Spatial filters: Gaussian smoothing is the typical choice for additive noise,but it smooths out edges as well as noise by

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weighting neighbours based on distance only [1]. The median filter takes the middle value of the pixels in a window,removing the influence of some extreme values on the overall result, which is why it is a common choice for impulse noise [1].
Tomasi and Manduchi added an intensity term to the Gaussian weight in the bilateral filter, so pixels on the far side of a strong edge count for very little [2]. Non-local means goes a step further and averages patches that look alike wherever they appear in the image [6], and BM3D groups similar patches and filters them together in a transfo domain, which became the reference point for hand-designed denoisers [7].

Subspace methods: PCA describes images using a few orthogonal directions that capture most of the variance [3]. White noise spreads its own energy equally in all directions, while the content of the image piles up in the first few, so keeping only the leading components improves the signal-to-noise ratio. The real challenge is how many to retain: too few leads to loss in detail, too many and the noise comes back.

Learned denoisers: Vincent et al. showed that an autoencoder trained to rebuild clean inputs from corrupted ones learns more robust features than one trained on clean data alone [4]. Convolutions layers, instead of dense layers, leverage local image structure [8].Mao et al. found that deeper encoder–decoder networks with skip connections are able to restore images even better [9], and DnCNN revised the task as residual learning, where the network predicts the noise rather than the clean image [5].

Quality measures: MSE and the PSNR which are derived from it are simple to calculate but do not consider image structure. Wang et al. introduced the structural similarity index (SSIM), which compares local luminance, contrast and structure which is closer to human judgement [10]. We report both, since a method can improve one without the other.

Most of this work targets large natural photographs and compares against a handful of rivals. Small, centred objects, impulse noise next to Gaussian noise, and the question of what happens when the input is only lightly corrupted get much less attention. This paper looks at exactly those cases, using a deliberately small experiment that anyone can rerun.

III. METHODOLOGY

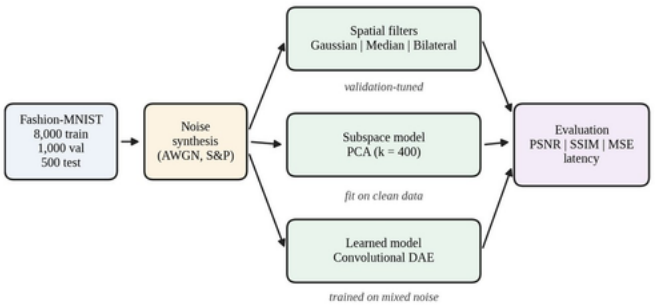


Fig. 1. Experimental pipeline: noisy test images are restored by three method families and scored with the same metrics.

A. Dataset and Noise Models

We use Fashion-MNIST [11], which has 28 × 28 grayscale images of ten clothing classes. It is a tougher test than handwritten digits, because garments have textured interiors and graded intensities rather than thin strokes on a blank background. Pixel values are scaled to [0,1]. PCA and the CDAE are fitted on the first 8,000 training images. Images 8,000 to 9,000 of the training set serve as a validation pool (the first 200 are used to tune the filters), and every reported number comes from the first 500 test images.
Each clean image x is observed as:

$$y = x + \eta \tag{1}$$

where, for additive white Gaussian noise (AWGN), $\eta \sim N(0, \sigma 2\ n)$ with $\sigma_n = \sigma/255$, and the result is clipped to [0, 1]. We use σ = 15 (mild), 25 (moderate) and 50 (severe). For salt-and-pepper (S&P) noise a fraction d of the pixels is overwritten, half with 1.0 and half with 0.0, using d = 0.05 and 0.10. The random seed is fixed at 42.

B. Classical Spatial Filters

The Gaussian, median and bilateral filters come from OpenCV. For each noise condition we chose the parameters by grid search on the 200 validation images, maximising mean PSNR. The grids were: Gaussian kernel size {3, 5} and spread {0.5, 0.8, 1.0, 1.5}; median kernel size {3, 5}; bilateral diameter {5, 7, 9}, range spread {0.05, 0.10, 0.15, 0.25} and spatial spread {1.5, 3.0}. The selected values are in Table II. This does give the filters an advantage, since they are tuned knowing the noise type and level, which the learned models are never told.

TABLE II
VALIDATION-SELECTED FILTER PARAMETERS

Condition	Gaussian (k, Median σ)	(k)	Bilateral (d, σ_c, σ_s)
AWGN σ=15	3, 0.5	3	7, 0.10, 1.5
AWGN σ=25	3, 0.5	3	5, 0.15, 3.0
AWGN σ=50	5, 0.5	3	7, 0.25, 3.0
S&P d=0.05	5, 0.5	3	9, 0.25, 3.0
S&P d=0.10	3, 0.8	3	9, 0.25, 3.0

C. PCA Subspace Projection

Treating the clean training images as 784-dimensional vectors, we compute the mean μ and the orthonormal eigenvectors U from their covariance matrix. A noisy vector is projected onto the leading k eigenvectors U_k and mapped back:

$$\hat{x} = \mu + U_k U_k^T (y - \mu) \tag{2}$$

With white noise of variance σ^2_n, the expected squared error is the energy in the discarded components plus about $k\sigma^2_n$ of retained noise, so a larger k cuts truncation error but lets more noise through. We chose k on validation images with σ = 25, trying {10, 20, 35, 50, 75, 82, 120, 181, 220, 300, 400, 500, 600}. Validation PSNR peaked at k = 400 (22.81 dB, 98.7% of the variance) and fell to 22.67 dB at k = 500. The same value was then used for all five conditions.

D. Convolutional Denoising Autoencoder

The CDAE has an encoder of two convolution–batch-norm –ReLU–max-pool stages (1→32 and 32→64 channels, 3 × 3 kernels), a 64-channel bottleneck convolution at 7 × 7 resolution, and a decoder of two transposed convolutions (stride 2, 64→32 and 32→16 channels) followed by a 3 × 3 convolution to one channel with a sigmoid output. Batch normalisation [16] follows every hidden convolution. The network has 66,593 trainable parameters and has no skip connections. It minimises the mean squared error between the restored output and the clean target:

$$L(\theta) = (1/B)\, \Sigma_i \, \| f_\theta(y_i) - x_i \|^2 \qquad (3)$$

The 8,000 training images are split among noise types as 25% each for AWGN σ = 15, 25 and 50, 15% for S&P d = 0.05 and 10% for d = 0.10, so one model handles every condition. Training used Adam [17] with learning rate 10^{-3}, weight decay 10^{-5} and a plateau scheduler (factor 0.5, patience 2) for 10 epochs. Implementation is in PyTorch [18] and PCA uses scikit-learn [20].

E. Evaluation Metrics

Restoration quality is scored by mean per-image MSE, PSNR and SSIM against the clean image, averaged over the 500 test images. PSNR is computed as

$$PSNR = 10 \log_{10}(1 / MSE)\ dB \qquad (4)$$

because the peak value is 1.0. SSIM [13] uses the default 7 × 7 window of scikit-image [21] with a data range of 1.0. Latency is the mean CPU time per image; the CDAE is run in batches of 128.

IV. RESULTS AND DISCUSSION

A. Experimental Setup

Everything ran on a multi-core CPU workstation with Python 3.12, PyTorch 2.11, OpenCV 5.0 and scikit-learn 1.8, and no GPU. CDAE training took 123.4 s. The training loss fell from 0.0393 to 0.0052 and the validation loss from 0.0119 to 0.0054 over the ten epochs (Fig. 5), and the two curves stayed close together. Fitting PCA took 0.45 s.

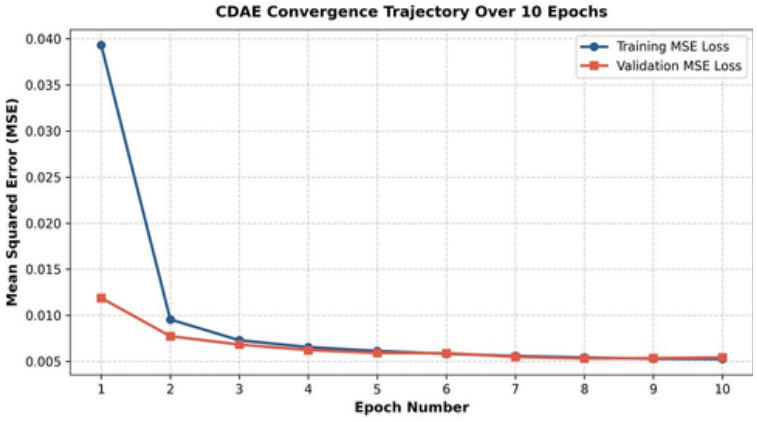


Fig. 5. CDAE training and validation MSE over ten epochs.

B. Quantitative Results

Table I and Fig. 3 hold the main numbers. Three things stand out.

Mild and moderate Gaussian noise. The tuned bilateral filter has the best PSNR at σ = 15 (28.35 dB, +2.26 dB over the noisy input) and at σ = 25 (24.52 dB, +2.74 dB). At σ = 25 the CDAE comes within 0.45 dB (24.07 dB), whereas at σ = 15 it sits 3.74 dB lower; in both cases its SSIM is higher (0.8566 vs. 0.8481, and 0.8447 vs. 0.7835). PCA gains little (+1.18 dB at σ = 25). The median filter blurs fine texture and has no outliers to remove here, so it ends up below the noisy input in both settings.

Severe Gaussian noise: At σ = 50 every method beats the 16.10 dB input, but the CDAE pulls well ahead with 21.95 dB. That is 2.76 dB better than the best filter (bilateral, 19.19 dB) and 5.85 dB better than the input. Its SSIM of 0.7731 is also far above the bilateral filter's 0.6521. Fig. 2 shows the reason: the filters either leave speckle behind or wipe out texture, while the CDAE returns a smooth garment with its outline intact.

Impulse noise: Among the classical methods the median filter is clearly the best (20.87 dB and 20.05 dB), which makes sense for outliers that sit at the extremes of the intensity range. The bilateral filter barely beats doing nothing (17.44 dB and 14.70 dB, against 17.08 dB and 14.10 dB for the input). A salt or pepper pixel differs so much from its neighbours that the range weight reads it as an edge and keeps it. The CDAE reaches 23.37 dB and 21.97 dB, which is 2.50 dB and 1.92 dB above the median filter, with the highest SSIM as well.

TABLE I
PSNR (DB) AND SSIM OF ALL METHODS ON 500 FASHION-MNIST TEST IMAGES. BEST VALUE AMONG DENOISERS IN BOLD

Method	AWGN σ=15		AWGN σ=25		AWGN σ=50		S&P d=0.05		S&P d=0.10	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
Noisy input	26.09	0.8049	21.78	0.7273	16.10	0.5796	17.08	0.7208	14.10	0.5877
Gaussian	24.86	0.8081	22.56	0.7480	18.18	0.6296	19.71	0.7469	17.80	0.6241
Median	21.04	0.7335	20.26	0.6787	18.12	0.5831	20.87	0.8107	20.05	0.7937
Bilateral	**28.35**	0.8481	**24.52**	0.7835	19.19	0.6521	17.44	0.6825	14.70	0.5759
PCA (k=400)	25.97	0.8054	22.96	0.7444	18.04	0.6214	19.86	0.7070	17.01	0.6160
CDAE	24.61	**0.8566**	24.07	**0.8447**	**21.95**	**0.7731**	**23.37**	**0.8360**	**21.97**	**0.8028**

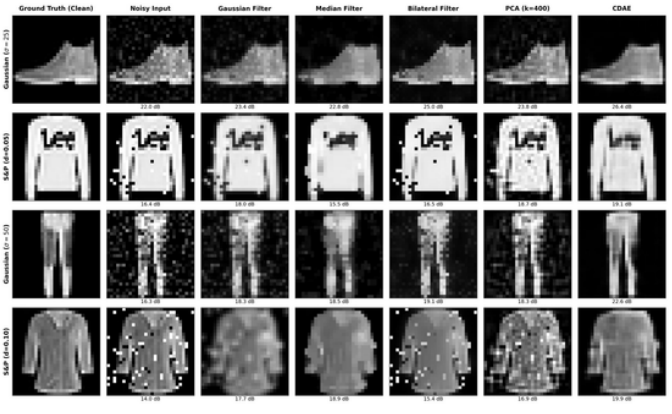


Fig. 2. Example restorations. Rows: AWGN σ = 25, S&P d = 0.05, AWGN σ = 50, S&P d = 0.10. Columns: clean, noisy, Gaussian, median, bilateral, PCA, CDAE. Per-image PSNR is printed under each image.

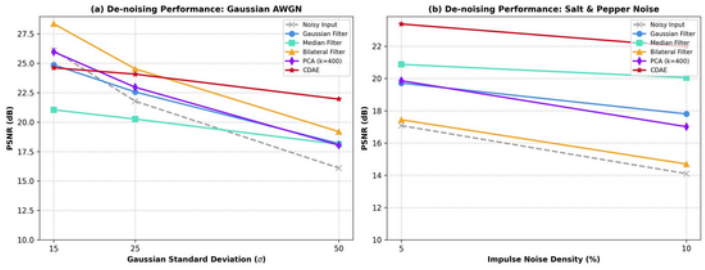


Fig. 3. Mean PSNR versus noise strength for (a) Gaussian noise and (b) salt-and-pepper noise.

C. When Denoising Hurts

At σ = 15 the noisy input already scores 26.09 dB, and four of the five denoisers land below it: median −5.05 dB, Gaussian −1.23 dB, CDAE −1.48 dB and PCA −0.12 dB (Table I). Only the bilateral filter gains. A fixed smoothing strength that suits heavier noise removes more signal than noise here. For the CDAE this probably comes from the training mixture, where three quarters of the images carry stronger corruption than σ = 15, although we did not test that separately. The CDAE still has the best SSIM at σ = 15, so it gives up a little pixel accuracy while keeping structure. That is a good reason to read PSNR and SSIM together.

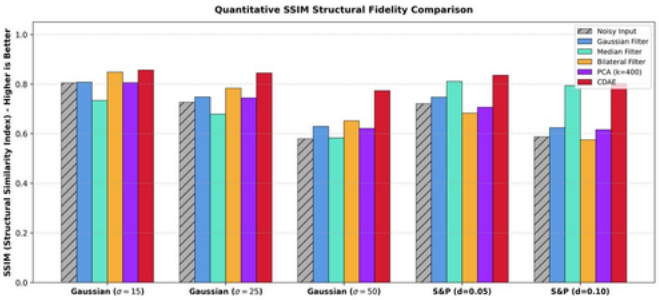


Fig. 4. SSIM of each method in the five noise conditions.

D. Computational Cost

Table III shows that the classical filters and PCA all take well under 0.1 ms per 28 × 28 image. The CDAE needs about 0.295 ms, roughly 5 to 60 times more, plus a one-off two- minute training run. That is still thousands of images per second on a CPU, so for images this small the cost is not a real obstacle. For large images or embedded hardware it would have to be measured again.

TABLE III
INFERENCE LATENCY AND FITTING COST

Method	Latency (ms/image)	Offline fitting
Gaussian filter	0.009	None
Median filter	0.005	None
Bilateral filter	0.055	None
PCA (k=400)	0.018	0.45 s
CDAE	0.295	123.4 s

E. Discussion and Limitations

The results suggest a simple rule of thumb. If the noise is known to be light and Gaussian, a tuned bilateral filter is hard to beat and needs no training. If it is impulsive, a median filter is a sound and cheap choice. If the noise type or level is unknown or strong, a single trained CDAE is the safer bet, because it gave the best SSIM everywhere and the best PSNR in three of five conditions without being told which noise it faced.

There are limits to what we can claim. The comparison is not symmetric: the filters were tuned for each condition, while PCA (with k chosen at σ = 25 only) and the CDAE are single blind models, which probably understates PCA at σ = 50 and under impulse noise. The data are small grayscale images with synthetic noise, whereas real sensor noise depends on the signal and is spatially correlated. Results come from one seed and 500 test images, so we give no confidence intervals and would not read much into gaps below about 0.5 dB. Finally, the CDAE is a small ten-epoch model, and stronger baselines such as BM3D [7] or DnCNN [5] were not included.

V. CONCLUSION

We compared three families of image denoisers on Fashion-MNIST under Gaussian and salt-and-pepper noise, and no single method won everywhere. A tuned bilateral filter gives the best PSNR for mild Gaussian noise, the median filter is the best classical option for impulses, and a compact 66,593-parameter CDAE trained on mixed noise gives the best SSIM in every condition and the best PSNR for strong

Gaussian and impulse noise, at a modest computational cost. The fact that most methods lowered PSNR on lightly noisy images is a useful reminder that denoising should only be applied when the noise level justifies it.

Next we plan to repeat the runs with several seeds and report confidence intervals, add residual-learning networks such as DnCNN and a BM3D baseline, and test on larger images with real noise, including how denoising affects downstream classification accuracy.

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